Supplementary Information

to "Reach, not speed, limits who benefits from AI testing tools and where vendors invest"

This appendix contains the proofs of the results in the main text (Section A), the pricing extensions (B), the investment extensions (C), the extensions on private information about reach and on mixed-complexity markets (D), the estimation machinery for the NHPP regression layer (E), the details of the public-data analyses (F), and a guide to the replication package (G). Every result is labeled **Proved**, **Proved under stated conditions**, or **Numerical**. Numerical results are computed at the illustrative parameters of Methods ($a = 20$, $b = 0.12$, $B = 1.6$, $c_0 = 0.8$, $k = 0.1$, $A_0 = 0.5$, $\ln \kappa \sim \text{Normal}(\ln 4.8, 0.9^2)$) and should be read as illustrations of mechanisms.

A. Proofs

A.1 Lemma 2 and Proposition 2 (adoption set)

Lemma A1. Let $g(x) = xh'(x)/h(x)$ for $x > 1$. Then $g(x) = \ln x/(x - 1 - \ln x)$, and g is strictly decreasing with $g(x) \to \infty$ as $x \downarrow 1$ and $g(x) \to 0$ as $x \to \infty$.

Proof. From $h(x) = 1 - (1 + \ln x)/x$ we get $h'(x) = \ln x/x^2$ and $h(x) = (x - 1 - \ln x)/x$, so g is as stated. Put $u = \ln x > 0$. Then $g = u/(e^{u} - 1 - u) = 1/\varphi(u)$ with $\varphi(u) = (e^{u} - 1 - u)/u$. We have $\varphi'(u) = [ue^{u} - e^{u} + 1]/u^2$. The numerator $n(u) = ue^{u} - e^{u} + 1$ has $n(0) = 0$ and $n'(u) = ue^{u} > 0$ for $u > 0$, so $n(u) > 0$ and φ is strictly increasing on $(0, \infty)$. Hence g is strictly decreasing. The limits follow from $\varphi(u) \to 0$ as $u \downarrow 0$ (since $e^{u} - 1 - u \approx u^2/2$) and $\varphi(u) \to \infty$ as $u \to \infty$. ∎

Proof of Lemma 2. For $\kappa > 1$ and $\rho > 1$, $R(\kappa) = h(\kappa)/h(\rho\kappa)$ is well defined and positive. Then $\mathrm{d} \ln R/\mathrm{d} \ln \kappa = g(\kappa) - g(\rho\kappa)$, which is positive since $\rho\kappa > \kappa$ and g is strictly decreasing. As $\kappa \downarrow 1$, $h(\kappa) \to 0$ while $h(\rho\kappa) \to h(\rho) > 0$, so $R \to 0$. As $\kappa \to \infty$, both h values tend to 1, so $R \to 1$. Finally, $\partial \ln R/\partial \rho = -\kappa h'(\rho\kappa)/h(\rho\kappa) = -g(\rho\kappa)/\rho < 0$. ∎

Proof of Proposition 2. If $\rho\kappa \le 1$ the AI saving is zero, and if $\kappa \le 1$ the human saving is zero. Hence for $\kappa \in (1/\rho, 1]$ the firm strictly prefers AI (saving positive against zero), and for $\kappa \le 1/\rho$ it is indifferent (both zero) and, by a tie-breaking convention, does not adopt. For $\kappa > 1$, adoption means $\Phi(\kappa) > 0$, that is, $A > R(\kappa)$. If $A \ge 1$ this holds for all $\kappa > 1$ because $R < 1$. If $A < 1$, Lemma 2 gives a unique $\kappa^* \in (1, \infty)$ with $R(\kappa^*) = A$, and adoption holds exactly for $\kappa < \kappa^*$. The comparative statics follow because R is increasing in κ (so κ^* increases when A rises, which

raises the level that R must reach), decreasing in ρ (so κ^* increases in ρ), and $\rho = AB/(c_0 + p)$ is increasing in A and B and decreasing in p. As $A \uparrow 1$, $R(\kappa^*) \to 1$, which requires $\kappa^* \to \infty$. ∎

Other regions. If $\rho \le 1$ and $A < 1$, the adoption set is empty (Proposition 5). If $\rho \le 1$ and $A > 1$, the adoption set is an upper half-line (κ_1, ∞) with $\kappa_1 > 1$; we use this region only in the proof of Proposition 7(iii). We do not use the case $\rho \le 1$, $A = 1$, in which $\Phi \le 0$ everywhere and adoption is empty.

A.2 Proposition 3 (general exclusion)

The proof is in the main text. Two remarks. First, the bound uses only that the firm's gross benefit under AI cannot exceed fAa, which holds whatever the mechanism. The threshold is therefore a sufficient condition, and the true exclusion boundary under a given mechanism is lower. Second, for the linear usage price, $S_L = fAa\, h(\rho\kappa) = fAa - c_L(1 + \ln(\rho\kappa))/(Bb)$, which is $fAa - O(\ln f)$ whenever $c_L = c_0 + p > 0$, and likewise $S_H = fa - O(\ln f)$, so $V/(fa) \to A - 1$.

A.3 Proposition 4 (scaling of stakes)

The proof is in the main text. The factor λ^* depends on the adoption set only through the geometric mean $\sqrt{(\kappa^*/\rho)}$ of its endpoints. At the illustrative parameters and $A = 0.9$, $\lambda^* = 0.485$ and the adopting share at the peak is 77 percent (Fig. 2 of the main text).

A.4 Propositions 5 and 6 (entry and the investment jump)

The proofs are in the main text. We add two remarks. (i) Continuity of Π^* in A (assumption P) is used for existence of the maximum in Proposition 6. We argue for it by dominated convergence: the adoption set and the profit integrand vary continuously with A and are dominated by an integrable function of κ (a multiple of $\ln \kappa$ on the adoption set). We have not written out the details, and for general stakes distributions the condition should be verified case by case. (ii) For the vendor's optimal price at a given A we use the supremum over prices; a maximum may not be attained, for instance at $A = 1$ where the optimal price approaches the adoption cliff from one side.

A.5 Proposition 7 (kink at parity)

Setup. Let $\ln \kappa \sim \text{Normal}(m, \sigma^2)$, with density q. For a given usage price, $\rho = AB/(c_0 + p)$, so $p - k = AB/\rho - c_0 - k$. The second-stage profit is $\pi = ((p - k)/(Bb))\, \mathrm{E}[\ln(\rho\kappa);\ \kappa \in S(\rho, A)]$, where S is the adoption set from Proposition 2 and the other regions of A.1. Write $P = B - c_0 - k$, $M = \mathrm{E}[\ln \kappa;\ \kappa > 1]/(Bb)$, $s = \Pr(\kappa > 1)$ and $C_{*\varepsilon*} = BM - Ps/(Bb)$, assumed positive (condition Q).

(i) Π^{}(1) = PM. *At* A = *1: if* ρ ≤ *1 then Φ ≤ 0 and no firm adopts. For* ρ = *1* + ε *with* ε > *0, the adoption set is (1/*ρ, ∞*) and* p − k = P − B ε*/(1* + ε*). Demand is E[ln(*ρκ*); κ > 1/*ρ*]/(*Bb*). The part of the integrand on (1/*ρ, *1] has ln* κ ≤ *0, and ln(*ρκ*) = ln* ρ + *ln* κ*, so demand is at most* M + *ln* ρ · *Pr(*κ > *1/*ρ*)/(*Bb*)* ≤ M + ε s*/(*Bb*)* + O*(*ε²*). Therefore* π ≤ PM − C_*{*ε*}* ε + O*(*ε²*). Under (Q) small* ε *is a loss. For large* ε *the comparison with* PM *is a global statement that we verified numerically (the optimum over* ρ *in the continuous-stakes model is at* ρ = *1.000000). As* ε ↓ *0, demand converges to* M *and the margin* p − k *to* P*, so profit approaches* PM *from below and Π^{}*(1) = *PM* as a supremum. In the illustration *PM* = 5.7729, matching the numerical optimum of the continuous model (5.7729).

(ii) Right derivative. Take $A = 1 + \eta$. *Lower bound:* set the price so that ρ is just above 1; all firms with $\kappa > 1$ adopt, and $p - k$ is $P + B\eta$ minus an arbitrarily small amount, so $\Pi^*(1 + \eta) \geq (P + B\eta)M$. *Upper bound:* extra gains are possible only from prices above the cliff, $\rho = 1 - \varepsilon'$ with $\varepsilon' > 0$. Then adoption requires $R(\kappa) < A$, and by the mean value theorem $\ln R(\kappa) = \varepsilon'\, g(\xi)$ with ξ between $\rho\kappa$ and κ; since g is decreasing, $\ln R(\kappa) \geq \varepsilon'\, g(\kappa)$. Since $\ln A \leq \eta$, adoption needs $g(\kappa) < \eta/\varepsilon'$, that is, $\kappa > \kappa_1$ with $g(\kappa_1) = \eta/\varepsilon'$. Near $\kappa = 1$, $g(\kappa) \approx 2/(\kappa - 1)$, so $\kappa_1 - 1 \approx 2\varepsilon'/\eta$. The lost demand $\int_1^{\kappa_1} \ln \kappa\, q(\kappa)\, d\kappa$ is bounded below by a constant times $(\varepsilon'/\eta)^2$ when ε'/η is small, and by a positive constant otherwise. The gain from the higher price is at most $BM\,\varepsilon'$. So the extra profit is at most $\max_{*\varepsilon*'} [BM\,\varepsilon' - P\,c(\varepsilon'/\eta)^2] = O(\eta^2)$. Combining with the lower bound, $\Pi^*(1 + \eta) = (P + B\eta)M + O(\eta^2)$, hence $\Pi^{*\prime}(1^+) = BM = \mathrm{E}[\ln \kappa;\, \kappa > 1]/b$. In the illustration this is 13.195. Numerically, at $\eta = 0.001$ and 0.004 the average slope is 13.21 and 13.20 with the optimal price at the cliff; at $\eta \geq 0.01$ the optimal ρ is slightly below 1 and the average slope exceeds BM by an amount of order η (13.5 to 14.2 for η between 0.01 and 0.04).

(iii) Left derivative. Take $A = 1 - \delta$ and a price with $\rho = 1 + \varepsilon$. Three facts are used. (a) The price margin is $p - k = P - B(\varepsilon + \delta)/(1 + \varepsilon)$ exactly. (b) The adoption set is contained in $(1/\rho, \bar{\kappa})$ with $\bar{\kappa} = g^{-1}(\delta/\varepsilon)$: adoption requires $\ln R < \ln A \leq -\delta$, while by the mean value theorem $\ln h(\rho\kappa) - \ln h(\kappa) = \ln \rho \cdot g(\xi)$ for some $\xi \in (\kappa, \rho\kappa)$. Adoption therefore needs $\ln \rho \cdot g(\xi) > \delta$, and since $g(\xi) \leq g(\kappa)$ we get $g(\kappa) > \delta/\ln \rho \geq \delta/\varepsilon$. (c) Demand is at most $M + \varepsilon\, s/(Bb) + C'\varepsilon^2 - Z(\bar{\kappa})$, where $Z(\bar{\kappa}) = \mathrm{E}[\ln \kappa;\, \kappa > \bar{\kappa}]/(Bb)$ is the excluded tail. Combining, the loss $L = PM - \pi$ satisfies, for small ε, $L \geq C_{*\varepsilon*}\,\varepsilon + BM\,\delta - C''\varepsilon^2 + (P/2)\, Z(\bar{\kappa}(\delta/\varepsilon))$. Suppose, for contradiction, that there are a constant K and a sequence $\delta_n \to 0$ with $L \leq K\delta_n$ at the optimal choice of ε_n. If ε_n stays above a fixed ε_0, L is bounded below by a positive constant, a contradiction. If $\varepsilon_n \to 0$, the first and third terms give ε_n

$\leq 2K\delta_n/C_{*\varepsilon*}$ for large n, so $\delta_n/\varepsilon_n \geq r_0 = C_{*\varepsilon*}/(2K) > 0$. Then $\kappa^- \leq g^{-1}(r_0) < \infty$ and $Z(\kappa^-) \geq Z(g^{-1}(r_0)) > 0$, so L is bounded below by a positive constant, again a contradiction with $L \leq K\delta_n \to 0$. Hence $(\Pi^*(1) - \Pi^*(1-\delta))/\delta \to \infty$. ∎

Caveats. The argument shows the logical structure fully but does not trace the uniformity of the constants C' and C'' in the $O(\varepsilon^2)$ terms. It also covers only the lognormal case. *Rate.* An approximation of the optimal discount gives $\ln(1/r(\delta)) \approx \sigma\sqrt{(2\ln(1/\delta))}$, so that the left slope grows like $BM + C_{*\varepsilon*}/r(\delta)$, which is slower than any power of $1/\delta$ and faster than any power of the logarithm. At the illustrative parameters the loss divided by δ is 47, 74, 94, 113, 139, 160, 183, 274, and 393 at $\delta = 0.1$, 0.04, 0.02, 0.01, 0.004, 0.002, 0.001, 10^{-4}, and 10^{-5}. The approximation overstates these numbers by about a factor of two at the smallest δ (991 versus 393) and should be used only for the qualitative form.

A.6 Proposition 8 (bunching)

Local maximum at parity. Let $\psi(A) = \Pi^*(A)/c(A)$ with $c(A) = (A - A_0)^2/2$. The sign of ψ' equals the sign of $\Pi^{*\prime}(A)(A - A_0)/2 - \Pi^*(A)$. As $A \uparrow 1$, $\Pi^{*\prime} \to \infty$ by Proposition 7(iii), so $\psi' > 0$ just below 1. As $A \downarrow 1$, $\Pi^{*\prime} \to BM$, and $BM(1 - A_0)/2 - PM < 0$ is equivalent to $B(1 - A_0) < 2P = 2(B - c_0 - k)$, that is, $B(1 + A_0) > 2(c_0 + k)$, which is condition (Q′). So ψ has a strict local maximum at 1, with value $\psi(1) = 2PM/(1 - A_0)^2 = 46.18$.

Global maximum and the interval. That ψ is largest at $A = 1$ among all $A > A_0$ is verified numerically ($\psi(1.1) = 40.5$, $\psi(1.3) = 36.7$). Given that, $\hat{\theta} = \psi(1)$ by Proposition 6, and $A^*(\theta) = 1$ requires $J(1) \geq J(A)$ for all A. For $A < 1$ this follows from the global maximum of ψ. For $A > 1$, it requires $\theta \geq [\Pi^*(A) - \Pi^*(1)]/[c(A) - c(1)]$, and taking the supremum over A gives θ_{low}. Letting $A \downarrow 1$ gives the local lower bound $BM/(1 - A_0) = 26.39$. The global value $\theta_{low} = 27.58$ (attained at $A = 1.09$) is numerical. In the continuous model the optimal reach is exactly 1.000 for $\theta = 30$, 35, 40, 45, and 46, and at $\theta = 27$ it moves to 1.12, as the bounds predict. An earlier grid-based calculation put the upper end at 45.4–45.9; this was an artifact of a coarse price grid that missed prices at the cliff, and the closed form 46.18 is correct. ∎

A.7 Proposition 9 (under-investment and subsidies)

Proof. Since $W \geq \Pi^*$, $\theta_s = \sup W/c \geq \sup \Pi^*/c = \hat{\theta}$. At A_t, some firms have positive net saving from adopting, so $CS(A_t) > 0$ and $W(A_t)/c(A_t) > \hat{\theta}$, which makes the inequality strict. Applying Proposition 6 to W shows that the planner's optimum jumps at θ_s. A subsidy at rate s

multiplies the vendor's cost by $(1 - s)$, replacing θ by $(1 - s)\theta$ in J, so the vendor's threshold becomes $\theta/(1 - s)$, which reaches θ_s when $s = 1 - \theta/\theta_s$. For smaller s, vendors with θ between $\theta/(1 - s)$ and θ_s do not invest although the planner would. ∎

Numerical values. With reach capped at 1.3, $\theta_s = 65.09$ and $s^* = 1 - 46.18/65.09 = 0.29$. Earlier calculations on a coarser grid gave $\theta_s = 61.5$ and $s^* = 0.25$. The cap matters because the maximum of W/c is at the cap; with a different cap the number changes. Consumer surplus is the sum over adopters of the net saving, computed as $\mathrm{CS}(A, \rho) = (1/b)\int \kappa\, \Phi^{+}(\kappa)\, q(\kappa)\, d\kappa$ at the vendor's optimal price.

A.8 Duopoly (main text)

Let $U(A_i, A_j)$ be the second-stage profit of vendor i. *Lemma A2:* $U(A_i, A_j) \le \Pi^*(A_i)$. *Proof.* At any price pair, the firms that choose vendor i are a subset of the firms that prefer i to human testing, and each such firm's testing time is the same as under monopoly at that price. Hence vendor i's demand is no greater than a monopolist's at the same price, and so is its profit. ∎ *Result 1.* If $\theta > \theta$, (A_0, A_0) is the unique pure-strategy equilibrium: any $A > A_0$ gives $U - \theta c \le \Pi^*(A) - \theta c(A) < 0$. *Result 2.* There is no symmetric equilibrium with $A_1 = A_2 > A_0$: with identical tools Bertrand competition drives the price to k and $U = 0$, so either vendor gains $\theta c(A)$ by deviating to A_0. This assumes continuous prices and the usual Bertrand tie-breaking. *Result 3.* Let A_1 be a monopoly optimum and $\theta_f(A_1) = \sup_{*A*>*A*_{0}\}} U(A, A_1)/c(A)$. If $\theta \ge \theta_f(A_1)$, then (A_1, A_0) is a pure-strategy equilibrium: given a follower at A_0, the leader's best response is A_1, and a follower deviation to A earns $U(A, A_1) - \theta c(A) \le (\theta_f - \theta)c(A) \le 0$. This condition is sufficient for a preemption equilibrium; we have not shown that such an equilibrium is the only one. Numerically (seven-point grid) $\theta_f(1.0) \approx 24$, so the preemption equilibrium with the leader at parity exists for θ between 24 and 46.

A.9 Summary of the status of the results

Table A1. Status of results

Result	Status	Where the gap lies
Lemma 1, Lemma 2, Proposition 2	Proved	Requires $\rho > 1$ (AI more effective than human per unit cost)
Proposition 3	Proved	A sufficient bound; actual exclusion is lower
Proposition 4	Proved	Lognormal stakes; adoption set endpoints fixed
Propositions 5, 6	Proved	Continuity of Π^* argued by

		dominated convergence, not written out
Proposition 7	Proved under (Q); lognormal	Uniformity of $O(\varepsilon^2)$ constants not traced; divergence rate approximate
Proposition 8	Local part proved; global part numerical	Global maximum at $A = 1$, $\theta_{low} = 27.58$
Proposition 9	Proved	Magnitudes depend on reach cap 1.3 and the welfare measure
Duopoly results	Proved (1, 2, 3 as stated)	Uniqueness of preemption not shown

B. Pricing Extensions: Committed-Spend Menus and Two-Part Tariffs

A linear usage price is a restricted instrument (Bergemann, Bonatti, and Smolin 2026). We compute what richer instruments yield in our setting. Firms have private stakes f. A firm that picks a contract with testing budget T and payment t obtains $fAa(1 - e^{-*BbT*}) - c_0T - t$, against an outside option equal to the human saving $S_H(f)$. The vendor's cost is kT. We solve the vendor's direct-mechanism problem as a linear program over a 60-point grid of stakes and a 40-point grid of budgets, allowing randomization, which gives an upper bound on the profit of any menu. We then solve for the best deterministic menu (one item per type) as a mixed-integer program, with optimality gap 0.0 percent, and for the best two-part tariff (fixed fee F, usage price p).

Table B1. Vendor profit under alternative pricing instruments (illustrative parameters)

Reach A	Linear price	Two-part tariff	Deterministic menu	Randomized upper bound	Menu / linear	Menu / bound
0.8	0.181	0.243	0.303	0.304	+68%	99.7%
0.9	1.066	1.340	1.719	1.729	+61%	99.4%
1.0	5.739	5.774	5.896	5.940	+3%	99.3%

Notes. Grid 60 × 40. The deterministic menu uses almost fully separating contracts (16 distinct items among 20 adopting types at $A = 0.9$), and the assigned budget increases with stakes. At a coarser grid (40 × 30) the gain at $A = 0.9$ is +56 percent, so magnitudes depend on the grid but the ordering (menu > two-part tariff > linear) does not.

Three conclusions. First, richer pricing is worth the most below parity (a 60 percent or larger gain in profit), and little at parity, where the linear price is nearly optimal. Second, a two-part tariff captures only 40 to 45 percent of the menu's gain. Third, and the one that matters for the main text, the top decile of firms by stakes adopts with probability zero at $A < 1$ and one at $A = 1$ under all three instruments, as Proposition 3 requires. Menus change how much the vendor earns from firms that adopt; they do not change who adopts at the top.

Mandated liability. Suppose the vendor must pay the firm a damages amount proportional to the firm's loss from failures of the tool, δf with verification probability φ. For

firms, the limit of $V/(fa)$ becomes $(A - 1) + \varphi\delta(1 - A)/a$, which is positive when $\varphi\delta > a$, so the firm-side exclusion disappears for large enough liability. The vendor's response is to reprice. At $A = 0.9$:

Table B2. Vendor response to mandated liability proportional to stakes

$\varphi\delta$	Vendor price	Vendor profit	Adopting share	Top-decile adoption
0	0.34	1.037	0.62	0.00
5	0.44	1.040	0.69	0.00
10	0.57	1.007	0.72	0.00
20	0.87	0.781	0.84	0.00
25	exit	0	0	0

Notes. Numerical, complete-information vendor, illustrative parameters.

Liability proportional to stakes raises adoption by moderate-stakes firms, but the vendor raises its price, the top decile stays out, and at high enough liability the vendor leaves the market. This is why we state Proposition 3 for bounded transfers. The only transfer that could reverse exclusion is one that scales with the firm's own stakes, and the vendor cannot sustain it.

Competition among tools of reach below parity. Suppose there are n tools with reach $A_j < 1$. For each tool, $V_j/(fa) \to A_j - 1 < 0$ as $f \to \infty$ (Proposition 3 applied to each), so for large enough stakes the human option beats every tool, whatever the prices. The exclusion of the top does not depend on the number of vendors or the intensity of price competition. With one tool of reach below parity and another at or above, the high-reach vendor serves the top segment, and the low-reach vendor takes some low-stakes firms. Numerically, with $A_1 = 0.6$ and $A_2 = 1.0$ the high-reach vendor's price stays near its monopoly price (0.78 versus 0.79).

C. Investment Extensions

C.1 Two vendors: payoff matrix and equilibria

Table C1 reports second-stage profits (before investment costs) of the row vendor against the column vendor, for a reach grid. Table C2 reports the pure-strategy equilibria of the investment game for a cost parameter θ on a seven-point reach grid {0.5, 0.7, 0.8, 0.9, 1.0, 1.1, 1.3}.

Table C1. Second-stage profits of the row vendor (own reach in rows, rival reach in columns)

Own \ rival	0.50	0.80	0.90	1.00	1.10	1.30
0.90	1.03	0.96	0.00	0.17	0.33	0.44
1.00	5.71	5.00	2.84	0.00	0.20	0.44

1.10	7.28	7.05	5.55	2.88	0.00	0.36
1.30	10.75	10.75	9.98	7.82	5.71	0.00

Notes. Numerical, coarse grid. Zeros on the diagonal reflect Bertrand competition at equal reach.

Table C2. Pure-strategy equilibria of the investment game

θ	Equilibrium (A_1, A_2)	Monopolist's choice	Planner's choice
46–200	(0.5, 0.5): no investment	0.5	(0.5, 1.3) for $\theta \le 60$
30–40	(1.0, 0.5)	1.0	(0.5, 1.3)
20	(1.3, 0.5)	1.3	(0.5, 1.3)
10	(1.3, 0.5)	1.3	(1.1, 1.3)

Notes. Numerical, seven-point grid; threshold position is approximate. The planner prefers one vendor to invest to the cap, since duplicate investment is wasteful, and two differentiated vendors only when θ is low.

In equilibrium at most one vendor invests, and the investment threshold is the monopolist's. The market is a natural monopoly in reach.

C.2 Learning from usage

Consider two periods with discount factor 0.9. The vendor invests in reach A_1 in period one and sets a first-period price. Reach in period two is $A_2 = \min(1.3, A_1 + \lambda \times \text{first-period adoption share})$, where λ measures learning intensity. Investment occurs only in period one.

Table C3. Learning and the investment threshold

Learning λ	Threshold θ to reach parity by period 2	Relative to $\lambda = 0$	At $\theta = 46$: A_1^*, p_1^*, A_2
0	82	baseline	1.22, 1.40, 1.22
0.15	101	+23%	1.16, 1.15, 1.29
0.30	119	+45%	1.02, 0.80, 1.30
0.60	146	+78%	1.00, 0.75, 1.30

Notes. Numerical. The $\lambda = 0$ threshold is 82 rather than 46 because period-two profit raises the benefit of investment by a factor 1 + 0.9.

Learning substitutes for up-front investment: the threshold rises by up to 78 percent, and at the same θ the first-period reach falls. The vendor also lowers its first-period price (from 1.40 to 0.75–0.80), a form of penetration pricing. There is a trap, however. For learning alone to carry the vendor to parity, the initial reach must be high enough for some firms to adopt: at least 0.88, 0.82, 0.76, and 0.72 for $\lambda = 0.15$, 0.30, 0.60, and 1.00. At the starting reach of 0.5 nobody adopts and learning never starts. The role of investment support shifts toward getting initial reach to a level at which learning begins. The model is a two-period sketch with learning linear in the adoption share and equal across users, and is not a full dynamic model.

C.3 Private reach and the role of liability as commitment

Suppose reach is the vendor's private information and the investment is unobservable. Under usage pricing alone, the firms' beliefs, not true reach, determine the vendor's revenue, so the vendor has no incentive to invest and stays at A_0 (moral hazard). A fixed-damages warranty D paid on verified failure (verification probability φ) gives the vendor a marginal benefit of reach, because expected liability $\varphi(1 - A)Ds$ falls in A, where s is the adopting share. With a quadratic cost the vendor's reach is $A = \min(1, A_0 + \varphi Ds/\theta)$, and reaching parity requires $D \geq D_{\min} = \theta(1 - A_0)/(\varphi s)$. The closed form is proved. The equilibrium structure is numerical. Because adoption itself depends on the beliefs about reach, there are multiple equilibria, one at parity and one with partial investment (reach 0.52 to 0.88 at the parameters we tried). Warranties, in this sense, are a necessary condition for investment under hidden action, and a reach beyond parity cannot be induced because the warranty's marginal benefit vanishes at $A = 1$ (the vendor never fails the test).

D. Private Information about Reach and Mixed-Complexity Markets

This section treats the case in which reach is the vendor's private information, with two types $A_L < A_H$ and a common prior. Firms observe contracts and form beliefs. The main message is a comparison of instruments. We do not claim novelty for the basic separation mechanisms, which follow the logic of warranty signaling (Bakshi, Kim, and Savva 2015; Bauer 2026); our contribution here is to apply them in a setting with heterogeneous stakes and an outside human option.

Proposition D1 (usage pricing cannot signal; proved). The vendor's usage revenue $(p - k)T^*(\bar{A}, p)$ depends on the firms' belief $\bar{A}$ and the price, and not on the vendor's true type. Both types therefore have the same preferences over (p, belief) and no price can separate them. At a prior of 0.5 the pooling price is 0.23, and the high type earns 3 percent of its full-information profit.

Proposition D2 (success fees; ratio proved, remainder numerical). Under outcome-based pricing, vendor revenue is proportional to true reach and costs kT. If $k = 0$, a low type mimicking the high type earns A_L/A_H times the high type's profit and no separating contract exists in markets where the low type cannot survive on its own. For $k > 0$, mimicking a high type induces long testing that the low type finds less valuable; the cost kT becomes the signaling cost. At $A_L = 0.6$, $A_H = 1.0$:

Table D1. Success-fee signaling and the vendor's marginal cost k

Vendor cost k	High type's price, full information	High type's price, separating	High-type profit: separating / full information
0.00	0.36	none feasible	—
0.02	0.36	0.02	0.10 / 4.43
0.05	0.36	0.05	0.25 / 4.14
0.10	0.37	0.11	0.67 / 3.68
0.20	0.39	0.23	1.45 / 2.78
0.40	0.45	0.45	1.17 / 1.17

Notes. Numerical, illustrative parameters.

Proposition D3 (fixed-damages warranty; proved). With a usage price p and a warranty D paid on a verified failure (probability φ), a firm that believes the vendor is a high type with reach 1 expects zero liability, so its adoption decision does not depend on D, and the revenue from usage is independent of the vendor's type. The low type's profit from mimicking is $\Pi(p) - (1 - A_L)\varphi D s(p)$, and the high type's is $\Pi(p) - (1 - A_H)\varphi D s(p)$. Separation requires $D \geq D_{min}(p) = [\Pi(p) - \Pi_L^{own}]/[(1 - A_L)\varphi s(p)]$, where Π_L^{own} is the low type's own full-information profit. The high type's profit net of the warranty is $\Pi(p) - [(1 - A_H)/(1 - A_L)][\Pi(p) - \Pi_L^{own}]$. The warranty cost to the high type is therefore the fraction $(1 - A_H)/(1 - A_L)$ of the mimicking surplus, and vanishes when $A_H = 1$. The result does not depend on the vendor's marginal cost k, which contrasts with Proposition D2.

Table D2. Fixed-damages warranty: minimum warranty and retained profit

A_H	k	D_{min} (φ = 0.25 / 1.0)	High-type profit / full information	Retention
1.0	0.0	68.4 / 17.1	6.564 / 6.564	100%
1.0	0.1	59.8 / 14.9	5.738 / 5.738	100%
1.0	0.4	34.0 / 8.5	3.258 / 3.258	100%
0.9	0.0	22.2 / 5.5	1.140 / 1.519	75%
0.9	0.1	16.4 / 4.1	0.778 / 1.037	75%
0.9	0.4	3.8 / 1.0	0.097 / 0.129	75%

Notes. Low type with $A_L = 0.6$ and negligible own profit. Numerical values of Π; retention equals $1 - (1 - A_H)/(1 - A_L)$ as in the formula.

The comparison of instruments is: usage price, cannot separate; success fee, separates only with $k > 0$, retaining about 18 percent of the high type's profit at $k = 0.1$; fixed-damages warranty, separates for any k, with retention 100 percent at $A_H = 1$. The warranty needs the usual conditions: the vendor can pay (limited liability caps D), failures are verifiable, and the firm cannot manipulate the verification. The last is more fragile here because a firm could deliberately underinvest in testing to induce a failure.

Mixed-complexity market. Suppose a share λ_c of firms have complex tasks on which the AI's relative reach is multiplied by $\delta < 1$, the human technology being unchanged, and the vendor cannot tell the task type. If the complex segment cannot yield positive profit at any price, every profit function scales by $(1 - \lambda_c)$, so the separating price and the signaling cost do not depend on λ_c: the market gets smaller but the contract is unchanged (Table D3, left). The complex segment is served only when δ is high enough. At the parameters used, the critical δ lies between 0.6 and 0.8. At $\delta = 0.8$ something interesting occurs: the signaling distortion lowers the high type's separating price enough that 36 percent of complex-task firms adopt, whereas under the full-information price and under pooling none do. The signaling cost thus has an inclusion by-product, but the vendor earns only 0.20 from that segment against 3.68 from the easy segment, so it is not a motive.

Table D3. Mixed-complexity market ($\lambda_c = 0.5$)

Complexity penalty δ	Adoption, easy / complex (separating)	Pooled outcome pricing	Pooled usage pricing
1.0	0.98 / 0.98	0.31 / 0.31	0.29 / 0.29
0.8	0.98 / 0.36	0.31 / 0.00	0.29 / 0.00
0.6	0.98 / 0.00	0.31 / 0.00	0.29 / 0.00
0.4	0.98 / 0.00	0.31 / 0.00	0.29 / 0.00

Notes. Numerical. Prior expected profit is ordered separating outcome contract > pooled outcome contract > pooled usage contract at every λ_c and δ.

The policy implication, if one takes the numbers seriously, is that exclusion of complex tasks is a reach problem and not a pricing problem: no pricing instrument brings in complex tasks once δ is below the critical value. We regard these extensions as illustrations. The complexity layer takes the stakes distribution to be the same in easy and complex tasks. If complex tasks also carry higher stakes, the exclusion in Proposition 3 compounds it.

E. Estimation Machinery for the NHPP Regression Layer

The paper's theory is self-contained, and the model of the Methods section does not require estimation. For the empirical work that the model suggests, we built and validated a hierarchical Bayesian count model that estimates the pool *a* and rate *b* of paired human and AI-assisted test campaigns, with covariates for the use of AI assistance and for task complexity, and that carries posterior uncertainty to the net value of AI testing. This section documents what has been validated, using simulated data only. No estimates in the main text come from this model.

Model. For project i and test condition $L \in \{0, 1\}$, the expected number of faults found in interval $(t_{*j*-1}, t_{*j*}]$ is $a_{*iL*}(e^{-*b*_\{*iL*\}t_{*j*-1}} - e^{-*b*_\{*iL*\}t_{*j*}})$, the increment of a Goel–Okumoto mean value function. We set $\ln a_{*iL*} = \beta_0 + \beta_{llm}L + \beta_{cx}x_{*i*} + \beta_{int}L\,x_{*i*} + u_{*i*}$ and $\ln b_{*iL*} = \gamma_0 + \gamma_{llm}L + \gamma_{cx}x_{*i*} + v_{*i*}$, where x_{*i*} is task complexity and u, v are project random effects. Counts are Poisson, or negative binomial (gamma-mixed Poisson) with dispersion α for overdispersion. The model is implemented in PyMC with NUTS sampling. From the posterior we compute the cost-minimizing budget T^* and the net value V of AI-assisted testing at three complexity levels, as in Lemma 1.

Calibration check. We simulated 60 independent datasets (30 projects, 20 intervals, paired design) from fixed true parameters, refit the model to each, and recorded whether the 89 percent central posterior interval covered the true value. Coverage across the ten model parameters ranged from 0.83 to 0.97 (mean about 0.89), and for the net value V at complexity −1, 0, and +1 it was 0.88, 0.93, and 0.87. One of the 60 fits reported divergent transitions. The posterior probability that $V > 0$ pointed to the correct sign in 93, 65, and 87 percent of replications at the three complexity levels; the middle figure is low because the true value is close to zero there. These checks establish that the model recovers the data-generating process when it is correctly specified, and they say nothing about performance on real data.

Sensitivity. We evaluated the posterior mean of V and $\Pr(V > 0)$ over a grid of the cost of a field fault (0.5 to 20) and the cost premium of AI testing (1.0 to 3.0 times human). The sign of V depends on the premium, on the field-fault cost, and on task complexity, but posterior intervals are wide: at the default economics (field-fault cost 3, premium 1.3) the 89 percent interval for V has width 10 to 20 depending on complexity. At several grid points the posterior mean has the opposite sign to the value implied by the data-generating parameters, within the interval. Poisson and negative-binomial fits give similar posterior means (for example $V = 4.49$ versus 4.61 at average complexity), and the Poisson model has slightly narrower intervals (10.35 versus 10.67), which is the cost of ignoring overdispersion. The point of the exercise is that the economic conclusion from a single study of this size would be imprecise, which is a reason to treat the structural results of the paper as the contribution and not the numerical value of V. The grid is in `sensitivity_results.csv`.

F. Public-Data Analyses

All data in this section are publicly available. Code and derived tables are in the replication package; the raw files are downloaded by the scripts and are not redistributed.

F.1 SWE-bench Verified and agent trajectories

SWE-bench Verified (500 software-engineering tasks from 12 Python repositories) provides a human-assigned difficulty label based on the time an engineer would need: under 15 minutes (194 tasks), 15 minutes to 1 hour (261), 1 to 4 hours (42), over 4 hours (3). A published single run of a reasoning model on 495 of these tasks gives the resolution rates in Table F1. We also analyzed 80,036 trajectories on 3,591 task instances from a public collection of coding-agent runs (Nebius), most of them from a 70-billion-parameter open-weight model, to see how the share of tasks solved depends on the number of attempts.

Table F1. Agent resolution rate by human fix time (SWE-bench Verified, one reasoning-model run)

Human fix time	Tasks	Resolved	Mean conversation turns
Under 15 minutes	193	39.9%	17.9
15 minutes to 1 hour	258	25.2%	20.0
1 to 4 hours	41	4.9%	22.0
Over 4 hours	3	0%	37.3
All	495	29.1%	

Notes. The result is a task-level resolution rate, not a fault-detection rate. The run dates from 2024.

Table F2. Heterogeneous-instance model fits to agent trajectories

Model	Instances	Per-attempt success	Solved at least once (12 attempts)	Asymptotic reach (89% interval)	Extrapolated at 100 attempts (zero-inflated / beta)
70B open-weight	3,591	16.7%	23.3%	0.55 (0.43–0.75)	32.4% / 34.1%
8B open-weight	528	15.1%	16.5%	0.32 (0.22–1.00)	26.7% / 32.7%
405B open-weight	100	25.9%	48.0%	0.50 (0.43–0.57)	50.0% / 69.6%

Notes. The zero-inflated beta-binomial lets a share of instances be unsolvable; the beta-binomial does not. Model sizes are not comparable because instance sets differ.

In a fixed set of 223 instances with at least twelve attempts, the cumulative solve rate rises from 12.5 percent (one attempt) to 27.4 percent (12 attempts). The logarithmic NHPP has a sum of squared errors of 0.00023 and the exponential one 0.00104. Of the instances, 68 percent are unsolved in all 12 attempts. Asymptotic reach is model-dependent, but solve rates within an operational budget of 100 attempts agree between the two models within about two percentage

points for the 70B model. We therefore define reach in applications as a function of budget, $A(K)$.

F.2 Public Jira trackers

We streamed the Public Jira Dataset (Montgomery, Lüders, and Maalej 2022; about 56 GB uncompressed) for seven trackers (Apache, Qt, MongoDB, Sakai, Hyperledger, Spring, and Atlassian's Jira) and extracted 876,957 bug reports. Each report was assigned to the latest dated affected release that preceded its creation (43 percent of reports, 373,609, could be assigned). A release–tracker unit with at least 25 assigned reports and an observation window of 180 to 360 days contributed a series of 30-day counts. We fitted constant-rate, Goel–Okumoto, and Musa–Okumoto NHPP models by Poisson maximum likelihood, with parameter bounds to avoid degenerate fits.

Table F3. NHPP fits to 3,333 releases in public Jira trackers

Quantity	Result
Releases / projects / trackers	3,333 / 237 / 7
Best by AIC: Goel–Okumoto / Musa–Okumoto / constant	2,660 (80%) / 521 (16%) / 152 (5%)
Median AIC(GO) − AIC(MO)	−10.2
Releases whose fitted pool is at most 3× observed faults	97% (3,219)
Median fitted rate *b* per 30 days; implied half-life	0.518; about 1.3 months (IQR 0.8–2.2)
Median share of observed bug reports in first 30 days	38%
Median reports per release	44 (IQR 32–72)
Log standard deviation across releases: pool *a*; rate *b*	0.69; 1.05
Correlation of ln *a* and ln *b*	−0.20
Share best fit by Goel–Okumoto, lowest tracker	Atlassian Jira, 64% (36% Musa–Okumoto)

Notes. Fault reports in the field are driven by user exposure and the release-attribution rule. The fitted pool and rate should be read as parameters of a reporting curve, not as the size of a fault pool and the speed of a test campaign.

The share of high-priority reports is uncorrelated with the fitted pool and rate (correlations of −0.06), which does not contradict the model's assumption that stakes are independent of (a, b), although priority is a coarse proxy for stakes. There is no AI-use signal in these data, so they give the human baseline only.

F.3 Adoption of an AI coding assistant: what could and could not be done

We obtained a public replication package of a study of repositories that adopted an AI coding assistant (813 adopting repositories, about 78,000 candidate repositories, monthly panels of issues, commits, and code-quality metrics). We used it to (i) screen repositories for use as a

difference-in-differences sample, and (ii) measure issue-label usage so that defect-related issues could be separated. Of 196 repositories that passed the screen, we collected issues through the GitHub GraphQL API for 191 (five repositories had been deleted or made private), obtaining 114,362 issues (56,677 before and 57,685 after adoption). A label was present on 71 percent of issues and a defect-like label on 32 percent. Between 53 and 69 repositories had enough defect-labeled issues both before and after adoption (at least 10 or 5, respectively) for any repository-level analysis. Among the 53, the mean monthly count of defect-labeled issues was 27.8 before and 45.9 after adoption, higher after in 37 repositories; there is no control group in this calculation and it is not evidence of any effect. The sample is too small for a controlled difference-in-differences design (Callaway and Sant'Anna 2021), and control-group labels were not collected.

A crude test of Proposition 2 using GitHub stars as a proxy for stakes gave U-shaped relative adoption intensity (a quadratic coefficient of 0.221 in a logistic regression, $p < 0.001$), with the largest ratios at the extremes of the popularity distribution. We do not interpret this as evidence against the model. Stars measure popularity and not the cost of faults; large projects attract more contributors, so someone is more likely to commit a configuration file; the control sample is not documented well enough to read ratios as probabilities; and low-star repositories are mostly personal projects, not firms. A proper test would need a better proxy for stakes (for example the number of dependents or production use), contributor-level adoption data, and a documented control design.

F.4 A proposed identification strategy

A firm-level test of the model's predictions would use (i) the staggered introduction of an AI testing service across teams or firms; (ii) data on stakes, such as the cost of production incidents; (iii) estimates of the reach of the tool on the firm's own fault classes, perhaps from a pre-adoption trial; and (iv) a heterogeneity-robust difference-in-differences estimator. The model predicts that treatment effects on adoption and escaped faults vary with stakes in the single-peaked pattern of Proposition 2, and that the pattern flattens as measured reach approaches parity. The test would require firm-internal data that are not available in public sources.

G. Replication Package

The replication package accompanies the submission (*replication_package.zip*). It contains the Python code for every numerical result and figure, the derived data tables, a script that reproduces the key numbers cited in the text, and instructions for downloading the public raw data. A README documents the environment (Python 3.13, NumPy, SciPy, pandas, Matplotlib, PyMC, ArviZ), the order in which scripts are run, approximate run times, and which results require the large downloads. The package contains no credentials. The scripts that query the GitHub API read an access token from an environment variable and the token is not included.

Table G1. Where each result is produced

Result	Script
Adoption set, profit, price, menus	vendor_model.py, vendor_menu.py, vendor_menu_det.py
Kink and bunching (Propositions 7, 8)	kink_asymptotics.py
Investment jump, subsidies (Propositions 6, 9)	vendor_investment.py, vendor_investment_policy.py
Liability, competition, signaling	vendor_strategic.py, vendor_signaling.py, vendor_signaling_complexity.py
Duopoly, learning, private reach	vendor_dynamics.py
Figures 1–5	make_figures.py
NHPP regression layer and calibration	nhpp_llm_econ.py, coverage_check.py, sensitivity.py
Public-data analyses	jira_extract_bugs.py, jira_nhpp.py, swe_agent_curve.py, swe_beta_model.py, pilot_cursor_package.py, cursor_adoption_by_popularity.py, collect_issue_labels_gql.py
Key-number check	reproduce_key_numbers.py